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Design and Validation of HindiSER:Speech Emotion Recognition Dataset for Hindi Language

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***Abstract*— Every human prefers to express their emotions in their mother tongue. In India, Hindi is considered the National Language and is spoken by around 45% of the population. The unavailability of emotional speech databases in many of the regional languages is a significant bottleneck in the domain of Human-Computer Interaction (HCI). To address this issue of Low-resource regional language, we have proposed a new Hindi language emotional speech database "HindiSER". This database has a total 520 emotional utterances recorded from 26 non-professional native Hindi speakers in the age group of 19-45 years. HindiSER has emotional utterances of happy, sad, neutral, and angry emotion classes of 5 emotionally neutral Hindi sentences. Thus, the number of utterances in each emotional class is 130. Objective validation is performed with Support Vector Machine (SVM) classifier, K-Nearest Neighbor (KNN) and also with Multi-Layer Perceptron (MLP) classifier. Subjective validation of HindiSER is done through a listening test. Performance comparison of these classifiers using spectral features shows better recall on the happy emotion class and lowest recall on neutral emotion.**

Keywords—Hindi Speech Emotion Recognition Database, Low Resource Regional language emotion database, MFCC, Spectral features, SVM, KNN, MLP

I. Introduction

In Human Computer Interaction (HCI), the machine should be able to understand, interoperate instructions given by the end user. Human speech is one of the most user friendly and easiest input to the machine. Along with input speech, the emotional state of user will definitely add more information and naturalness in today's cognitive intelligence world. Therefore, the detection of affective state is a more demanding domain in HCI.[1] Additionally, input speech in the regional language 'Hindi' will be preferred. According to 2021 Census of India Hindi language is the most spoken followed by Bangala and Marathi as represented in Fig.1.

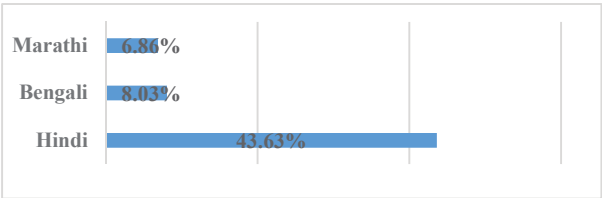

Fig.1. (%) of First Language Speakers (2011 census of India)

Many of the researchers[2]–[5] initiated the experimentation on speech based Hindi language speech emotion recognition. For experimentation, Hindi emotional speech database is the crucial part in this work. According to Ayadi et.al. [6], identification of speakers and selection of sentences are major stages in creation of emotion speech corpus. Majority researchers [2],[5] preferred emotionally invariant sentences over emotional influenced one [3].

II. Literature Survey

Very few emotional speech datasets are designed by researchers to explore Hindi speech-based emotion recognition. Koolgudi et.al. [2] designed an acted Hindi emotional speech database for eight emotion classes. Table I. compares the proposed HindiSER with other available Hindi emotional speech datasets.

III. Methodology

A new HindiSER emotional speech database is proposed in this study. The entire database design is thoroughly explained with a description of recordings. The database is validated with traditional machine learning classifier: Support vector machine (SVM) and neural network Multi-layer perceptron (MLP) with different cross-validation methods as described in the subsequent sections.

A. HindiSER Datbase Design

A set of five sentences were selected to capture different linguistic aspects relevant to the Hindi language. Four prominent emotional classes are identified for recording. Speakers are students of art and are trained to utter sentences in different emotions. Training is done with proper awareness objective of database and presenting recordings uttered by experts and other available Hindi databases [2,5]. The HindiSER dataset provides a valuable resource for training and evaluating speech emotion recognition models designed explicitly for the Hindi language. It facilitates advancements in emotion recognition technology and its applications in various domains.

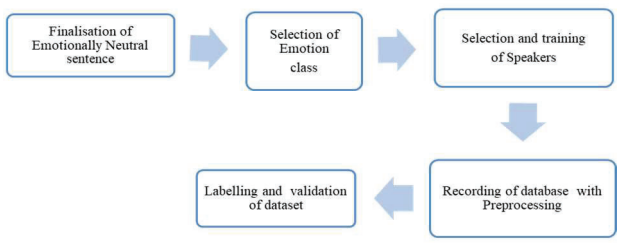

Fig.2 Stages of HindiSER database design

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TABLE I

COMPARISION WITH EXISTING HINDI LANGUAGE SPEECH EMOTION DATABASE

Ref. No. and Name of Database /Authors	[2] **IITKGP-SEHSC**	[3] **Bansal, Sweeta and Amita Dev et.al**	[4] **Akshat Agrawal & Anurag Jain et.al**	[5] **MNITJ-SEHSD**	**Proposed HindiSER**
# total audio records	12000	6048	500	500	520
# of speech-audio emotions	8	6	5	5	4
# of target sentences	15	12	25	10	5
Type of sentences	Emotionally Neutral	Emotionally Rich	Not Mentioned	Emotionally Neutral	Emotionally Neutral
# of participating actors	10 (5 Male And 5 Female)	28 (14 Male And 14 Female	4	10 (5 Male And 5 Female)	26 ((13 Male And 13 Female)
# Age Group(years)	28-48	20-30	Not Mentioned	19-27	19-45
# of sessions	10	3	1	1	1
Professional actors	Y	N	N	N	N
# No. of channels	Single	Stereo	Single	Mono	Stereo
Sampling Rate	16 KHz	44.1 KHz	44.1 KHz	Not Mentioned	48 KHz
Recording Environment	Sound Proof Room	Uncontrolled	Sound Proof Room	Not Mentioned	Uncontrolled
Name of Classes	Anger, Disgust, Fear, Happy, Neutral, Sad, Sarcastic And Surprise	Neutral, Happy, Anger, Sad, Sarcastic, Surprise	Anger, Happiness, Neutral, Sad, Surprise	Neutral, Angry, Happy, Fear And Sad.	Angry, Happy, Neutral Sad
Gender balance	Y	Y	Not Mentioned	Y	Y
Language	Hindi	Hindi	Hindi	Hindi	Hindi

Fig. 2 represents the stages used in designing of HindiSER. This database includes recordings from speakers of age group of 19-45 years, consisting of 13 male and 13 female non-professional speakers. Each speaker recorded five emotionally neutral sentences in four different emotions using a cell phone voice recorder in stereo, 32-bit resolution, a sampling rate of 48 KHz under ordinary room conditions. The recordings are subjected to pre-processing for background noise removal using the most recommended open-source platform of Audacity. With the help of majority voting, each sentence recording is identified and labeled carefully with the specific file name. E.g., an audio wave with file name 10-4-2 represents the second sentence in neutral emotion uttered by actor 10. Thus, each file name is assigned a unique label as per the details of labeling defined in Table II. As HindiSER is a Hindi speech-audio dataset developed for Speech Emotion Recognition (SER) in the Hindi language, the dataset comprises 520 speech-audio recordings featuring four prominent emotional states: Happy, Sad, Angry and Neutral. The emotions conveyed in the dataset are perceived as more natural and authentic [6]. This database is balanced in gender and number of recordings per emotion class, as shown in Fig. 3.

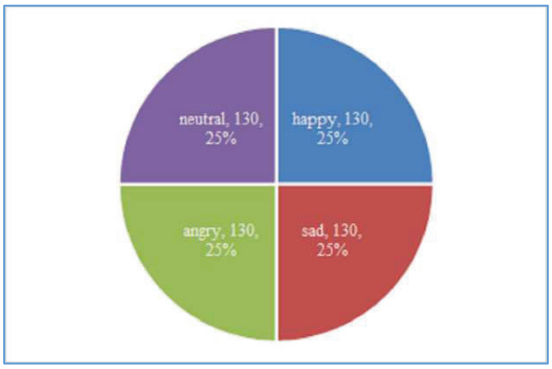

Fig.3. Emotion wise audio-recordings in HindiSER

B. *HindiSER Database Description*

HindiSER database has a total 520 emotional utterances from different users. Fig. 4.a to Fig. 4.d. represents the spectrogram and pitch waveform of female speaker 35 of sentence number one "कल छुट्टी है" in four emotions happy, angry sad and neutral, respectively. Table III represents emotion-wise average duration and pitch.

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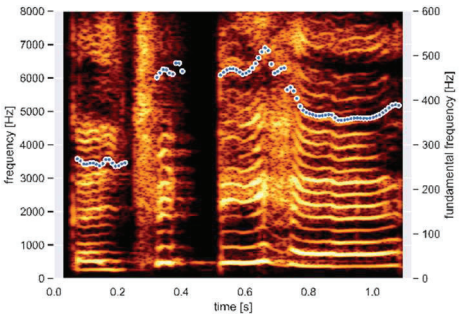
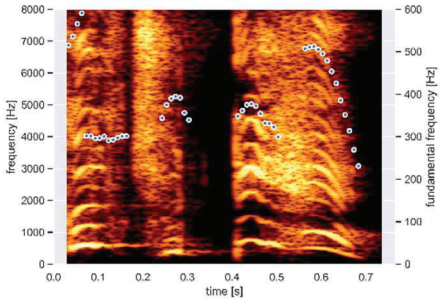
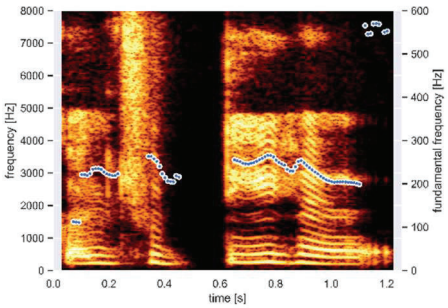


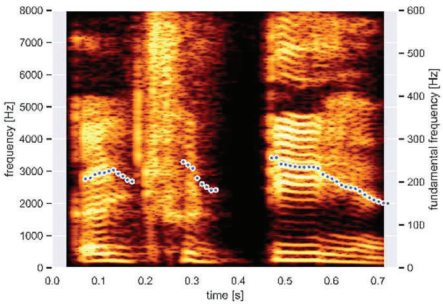
4.a. spectrogram and pitch of 35-1-1



4.b. spectrogram and pitch of 35-3-1



4.c. spectrogram and pitch of 35-2-1



4.d. spectrogram and pitch of 35-4-1

Fig.4.a.-4.d. Spectrogram and pitch waveforms of four distinct emotions of Female Speaker-35

A set of five representative sentences were chosen. These sentences were selected to capture different linguistic aspects and emotional content relevant to the Hindi language.

TABLE II

DETAILS OF UNIQUE IDENTIFIERS USED IN AUDIO FILE NAMES

Identifier (Count)	**Used in Unique File name**
Actor (26)	Male Speaker No.- 3,5,12,13,14,16,20,21,29,32,33,34,38 Female Speaker No -1,2,4,7,8,10,18,22,24,27,35,36,37
Emotion (4)	01 = Happy, 02 = Sad, 03 = Angry, 04 = Neutral
Statements (5)	1. “कल छुट्टी है । " (Tomorrow is a holiday) 2. "आज फिरसे देर हो गयी ।" (We got late again today) 3. "बाहर बारिश हो रही है ।" (It's raining outside) 4."मै रूम बदलणे वाला हु। "(I am going to change room) 5."आज हम बाहर जायेंगे। " (Today we will go out)

C. *Feature Extraction*

Spectral features derived from mapping with Mel scale i.e. Mel Frequency Cepstral Coefficients (MFCC) are very effective in detecting emotions [4],[7],[8],[9]. Many researchers also derived spectral features mapped with Gammatone filter [7] and MFCC derived from discrete wavelet transform (DWT)[8] for improvement in emotion recognition rate. Vishal B. Waghmare et. al [10] used MFCC features extracted from different vowels and consonants of Indo-Aryan and regional language Marathi to detect emotions through speech. The overview of all Indo-Aryan language experimentation with details of the database created is well elaborated by the authors [11].

With the librosa library [12], the mean of spectral features such as Mel Frequency spectral coefficients(MFCC), spectral centroid(SC), spectral roll-off (SR), zero crossing rate(ZCR) are extracted. Fig. 5. represents gender-wise variation of zcr_mean for all four emotion classes.

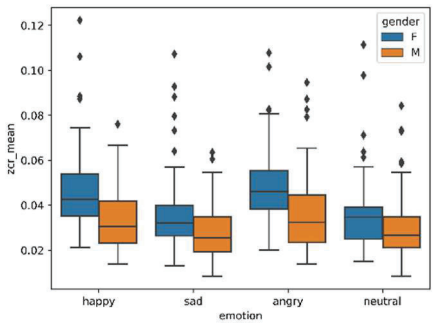


Fig. 5. Emotion-wise and gender-wise distribution of zcr_mean,

From HindiSER, emotion-wise pitch (F0) and the average duration is listed in Table III, which depicts that sad emotion has a maximum average duration of 1.38 seconds. In contrast, angry emotion has minimum duration of 1.12 seconds. The highest average pitch of 277.80 Hz is observed for happy emotion. To study variation in pitch for different emotions, pitch contour of sentence_01 "कल छुट्टी है” utterances in all different emotions is obtained for female speaker_01 and male speaker_21 receptively in Fig. 6.a. and Fig. 6.b

TABLE III

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EMOTION-WISE PITCH AND DURATION

Emotion	Average Duration(Seconds)	Pitch (Hz)
Happy	1.36 sec	277.80
Angry	1.12 sec	263.05
Neutral	1.23 sec	230.44
Sad	1.38 sec	231.03

In Fig. 6. the pitch contour clearly distinguishes different emotions for males as well as females.

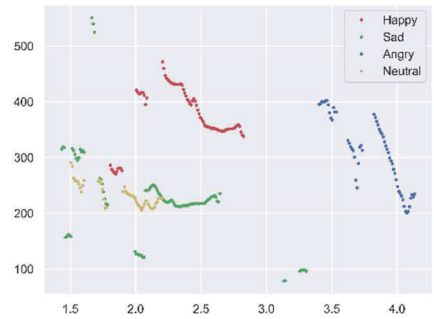

Fig.6. a.

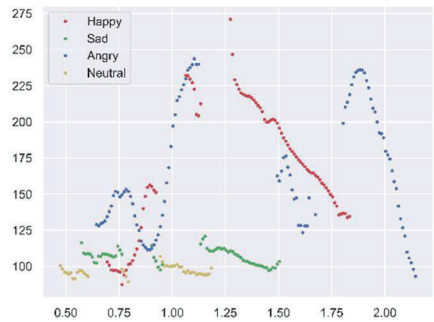

Fig.6. b.

Fig.6. Pitch Contour of sentence_01"कल छुट्टी है" for all four emotions of (a) Female Speaker_01 (b) Male Speaker_21

D. *Classification*

Extracted feature set of HindiSER is validated with machine learning classifiers Support Vector Machine (SVM), K-Neighorest Neighbour (KNN) and also with Multi Layer Perceptron (MLP). The corresponding specification of grid search obtained hyperparameters are listed in Table IV.

TABLE IV

SPECIFICATIONS OF CLASSIFIER USED

Classifier	**Gird search obtained Hyper parameters**
KNN	N=1
SVM	C=10, gamma=0.1, Rbf kernel
MLP	alpha=0.05, batch_size=256, epsilon=1e-08, Adaptive Learning rate hidden Layer=300,350,400

IV. DATASET EVALUATION

HindiSER is validated with machine learning classifiers as well as listening test.

A. *Subjective Evaluation*

Subjective evaluation is carried out by ten different subjects, through listening test [5]. Individual validation of all subjects is averaged and the average accuracy of 63% is achieved for HindiSER.

B. *Objective Evaluation*

Many researchers preferred support vector machine classifier in one vs all mode for objective validation of database[2], [3], [9].

Classification is done using 5-fold cross validation method with 80:20 as training and testing data distribution. The average accuracy achieved with SVM classifier is 61.53%, while that using MLP classifier is 65.54% and using KNN classifier average accuracy obtained is 60.57%. Fig.7(a), (b) and (c) depicts confusion matrix of experimentation of HindiSER database with KNN classifier, SVM classifier and MLP Classifier respectively. As observed from Fig. 8, happy emotion has highest recall of 74% with SVM and MLP classifier. For sad emotion, KNN and MLP classifier has highest recall of 70% whereas angry emotion has least recall as compared with all other emotional classes.

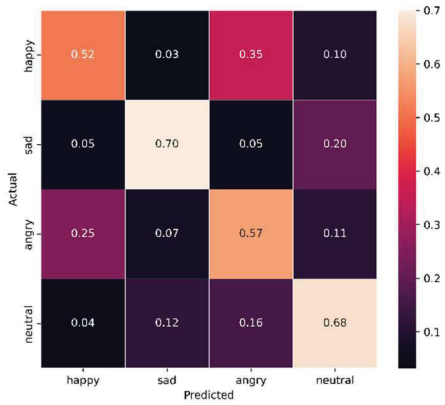

Fig.7.a.

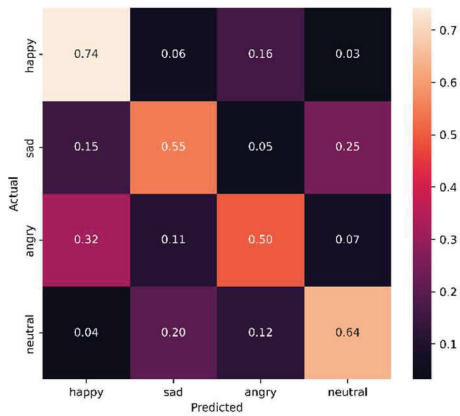

Fig.7.b.

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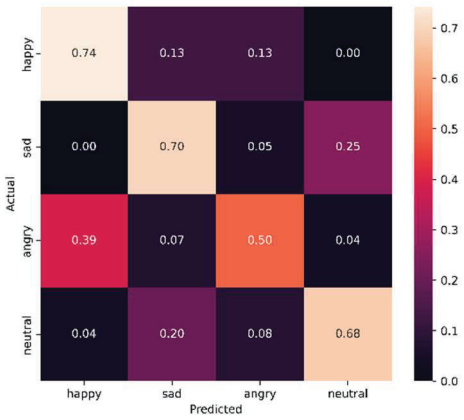

Fig.7.c

Fig.7. Confusion matrix of (a)KNN classifier (b) SVM Classifier and (c) MLP classifier with HindiSER

From Fig.7. a), b) and c) it is clear that, most misclassifications are angry emotion. Mostly anger is misclassified as happy, due to high pitch. In this experiment, with KNN classifier, misclassifications are less as compared with SVM and MLP classifier.

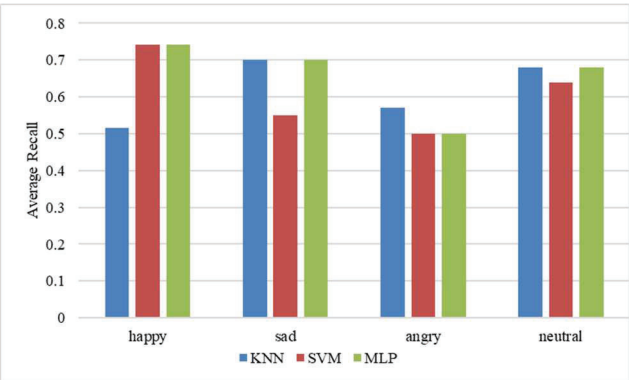

Fig.8. Performance Comparison of SVM, MLP and KNN classifier

V. Conclusion

Low resource Hindi language emotional speech dataset "HindiSER" is designed and validated in this study. The dataset comprises total of 520 audio recordings of 26 Hindi speakers who uttered five sentences in four different strong emotions such as happy, angry, neutral, and sad. All audio files are recorded in stereo mode, with a 32-bit resolution 48 KHz sampling rate in .wav format using the mobile phone. This new dataset is systematically designed, annotated and validated using traditional machine learning classifiers such as K-Nearest Neighbour, Support Vector Machine and Multi-Layer Perceptron. Experiment on HindiSER depicts happy emotion has the highest recognition recall, whereas neutral has the lowest average recall. The HindiSER is also validated through a listening test conducted by ten subjects with a validation accuracy of 63%. This database provides an opportunity to carry out experiments on regional language Hindi.

HindiSER database can be made available to the interested research scholars on request. This database can be used for machine learning, deep learning based applications of emotion recognition through speech. In the future, the HindiSER database will be expanded by adding more Hindi Native speaker's utterances and more trials with additional emotions.

Acknowledgment

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References

[1] Shinde, Ashwini S. and Patil,V.V. "Speech Emotion Recognition System: A Review," *Proc. 4th Int. Conf. Adv. Sci. Technol.*, 2021, doi: http://dx.doi.org/10.2139/ssrn.3869462.

[2] S. G. Koolagudi, R. Reddy, J. Yadav, and K. S. Rao, "IITKGP-SEHSC : Hindi speech corpus for emotion analysis," in *2011 International Conference on Devices and Communications, ICDeCom 2011 - Proceedings*, 2011. doi: 10.1109/ICDECOM.2011.5738540.

[3] S. Bansal and A. Dev, "Emotional hindi speech database," in *2013 International Conference Oriental COCOSDA held jointly with 2013 Conference on Asian Spoken Language Research and Evaluation (O-COCOSDA/CASLRE)*, 2013, pp. 1–4. doi: 10.1109/ICSDA.2013.6709867.

[4] A. Agrawal and A. Jain, "Speech emotion recognition of Hindi speech using statistical and machine learning techniques," *J. Interdiscip. Math.*, vol. 23, pp. 311–319, 2020, [Online]. Available: https://api.semanticscholar.org/CorpusID:219093614

[5] K. Chauhan, K. K. Sharma, and T. Varma, "MNITJ-SEHSD: A Hindi Emotional Speech Database," in *2023 International Conference on Communication, Circuits, and Systems (IC3S)*, 2023, pp. 1–6. doi: 10.1109/IC3S57698.2023.10169497.

[6] M. El Ayadi, M. S. Kamel, and F. Karray, "Survey on speech emotion recognition: Features, classification schemes, and databases," *Pattern Recognit.*, vol. 44, no. 3, pp. 572–587, Mar. 2011, doi: 10.1016/j.patcog.2010.09.020.

[7] S. Kumar and J. Yadav, "Emotion Recognition in Hindi Language using Gender Information, GMFCC, DMFCC and Deep LSTM," *J. Phys. Conf. Ser.*, vol. 1950, no. 1, p. 012049, Aug. 2021, doi: 10.1088/1742-6596/1950/1/012049.

[8] A. Bhavan, P. Chauhan, Hitkul, and R. R. Shah, "Bagged support vector machines for emotion recognition from speech," *Knowledge-Based Syst.*, vol. 184, Nov. 2019, doi: 10.1016/j.knosys.2019.104886.

[9] K. Chauhan, K. K. Sharma, and T. Varma, "A method for simplifying the spoken emotion recognition system using a shallow neural network and temporal feature stacking & pooling (TFSP)," *Multimed. Tools Appl.*, vol. 82, no. 8, pp. 11265–11283, 2023, doi: 10.1007/s11042-022-13463-1.

[10] V. B. Waghmare, R. R. Deshmukh, P. Shrishrimal, G. B. Janvale, and B. Ambedkar, "Emotion Recognition System from Artificial Marathi Speech using MFCC and LDA Techniques," 2014. [Online].https://api.semanticscholar.org/CorpusID:14211082

[11] S. T. Alam Monisha and S. Sultana, "A Review of the Advancement in Speech Emotion Recognition for Indo-Aryan and Dravidian Languages," *Adv. Human-Computer Interact.*, vol. 2022, p. 9602429, 2022, doi: 10.1155/2022/9602429.

[12] B. McFee *et al.*, "librosa: Audio and Music Signal Analysis in Python," in *SciPy*, 2015.

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